

Photo Album

A sarcoma is most likely to be diagnosed in which of the following patients?

- a. A 55 year old female with massive ascites and multiple peritoneal metastasis
- b. A 35 year old female with a left breast mass and enlarged axillary lymphnode
- c. A 25 year old male with an enlarged left testis
- d. A 15 year old male with a mass in the left femur, liver and lung metastasis
- e. A 5 year old male with right renal mass

A simple bacterial test for mutagenic carcinogen is:

- Redox test
- Ames test
- Gene splicing test
- Catalase test
- Bacteriophage test

A study of malignant neoplasms reveals that some of them have a greater potential for invasion and metastasis. Analysis of the cells of cancers that have metastasized shows a common change results in decreased cell surface expression of E-cadherin. Which of the following genes is most likely involved to produce these findings?

- a. Cyclin D
- b. BRCA-2
- c. Beta-catenin
- d. NF-1
- e. RAS

A surgical pathology report indicates that a certain neoplasm is graded as I on a scale of I to IV. Clearly, this neoplasm is stage I. These findings imply that this neoplasm:

- a. Has an in situ component
- b. Is well-differentiated and localized
- c. May spread via lymphatics
- d. Has probably arisen from epithelium
- e. Is not malignant

All are the features of MEN 2 (Multiple endocrine neoplasia type 2) syndrome, EXCEPT:

- a. Gastrinoma
- b. Medullary carcinoma of the thyroid
- c. Parathyroid hyperplasia
- d. RET proto-oncogene mutation
- e. Pheochromocytoma

An experiment is conducted in which proliferating cells are subjected to ionizing radiation. The ionizing radiation leads to arrest in a checkpoint that monitors completion of DNA replication. It is observed that there are increased numbers of chromosomal abnormalities in these cells. Which of the following is the checkpoint affected by this arrest? (completion of DNA replication is at the end of S phase)

- a. G1/S
- b. G0/G1
- c. S/G2
- d. M/G0
- e. G2/M

Anti-angiogenesis drugs act on malignant tumor by:

Anti-angiogenesis drugs act on malignant tumor by:

- a. Reduce blood flow and nutrient supply to tumor cells
- b. Promoting the immune response and removal of tumor cells
- c. Blocking hormonal stimulation of tumor cells
- d. Transport radioisotopes into tumor to destroy cells
- e. Transport chemotherapies into tumor to destroy cells

Biopsy of the bladder is performed in a 62 year old male who has complained of pain on urination. Microscopically, the epithelium shows cells with marked hyperchromatism and increased nuclear/cytoplasmic ratio involving the full thickness. However, these changes are confined to the epithelium above the basement membrane. This process is best described as:

- a. Minimal dysplasia
- b. Microinvasion
- c. Carcinoma in situ
- d. Hyperplasia
- e. Metaplasia

BRCA-1:

- a. Is associated with male breast cancer
- b. Is associated with breast and ovarian cancer
- c. Is associated with sporadic breast cancer
- d. Produces a truncated protein
- e. Is a type of oncogene

c-KIT mutations are seen in:

- a. Small-cell carcinoma of lung
- b. Neuroblastoma
- c. Ovarian cancers
- d. Gastrointestinal stromal tumors
- e. Skin cancers

Co-administration of 6-Mercaptopurine with Allopurinol may result in:

- a. Toxicity from 6-MP unless dose of 6-MP is decreased
- b. Toxicity from allopurinol unless dose of allopurinol is decreased
- c. Synergistic drug- drug interaction
- d. Acute gout exacerbation as a result of hyperuricemia
- e. Activation of 6-MP

Colorectal cancer is one of the most common cancers in Ethiopia. Which genes are most often mutated in hereditary non-polyposis colon cancer (Lynch syndrome)? *Correct answer: A*

- a. Cell cycle regulation
- b. DNA mismatch repair
- c. ER stress and apoptosis regulation
- d. Maintaining epithelial structure

Epithelial cells from colonic adenocarcinoma are shown to have a mutation that blocks hydrolysis of GTP-bound active RAS. Which of the following signaling pathways is most likely abnormally stimulated in the carcinoma cells?

- a. ADP
- b. p53
- c. MAP kinase
- d. Cyclic AMP
- e. BCR-ABL

Genes implicated in the pathogenesis of colorectal cancer include all of the following EXCEPT:

- a. K-ras gene
- b. BRCA1 gene
- c. APC gene
- d. c-myc gene
- e. P53 gene

Identify the correct statement.

- a. Infectious agents have little to do with carcinogenesis resulted from Host cell DNA damage due to chronic inflammation and oxidative stress.
- b. Besides being a predisposing factor for acquisition of Kaposi's Sarcoma, HIV per se the is one of the primary causes of esophageal cancer.
- c. A human adult may harbor a good number of microbial genes from the normal microbiota but not to the extent of having more microbial genes than that of human genome.
- d. Infectious agents can be utilized in the fight against cancer.
- e. The association of infectious agents with carcinogenesis can be linked to their roles in triggering deleterious host inflammatory response but not due to their role in the host immunity.

Identify the wrong statement concerning clinical manifestation of cancer

- a. Tumor Necrosis Factor (TNF) is largely responsible for cancer Cachexia (Emaciation)
- b. In the earlier stages as proliferating cells accumulate, cancers are always asymptomatic
- c. Nearly all symptoms that can be caused by cancer can also be caused more commonly by noncancerous diseases
- d. Paraneoplastic syndrome can be the earliest manifestation neoplasm
- e. In general loss of blood appears to be the main mechanism of anemia in cancers

If you find a 10 years old male child having vomiting, right lower abdominal pain of 2 months duration and if there is a clinically palpable mass which of the following imaging modality assesses nature and extent of the mass?

- a. Abdominal Ultrasound
- b. Abdominal CT scan
- c. Abdominal MRI
- d. None

In a female patient presenting with a swelling on the RT breast for 1 month, which physical characteristic is suggestive a malignant tumor?

- a. Mobile mass
- b. Painful mass
- c. Mass fixed to the underlying pectoralis major muscle
- d. All suggest malignancy

In an experiment, biologic characteristics of neoplastic and non-neoplastic cells are analyzed in culture. It is observed that cell division in cells derived from malignant neoplasms, but not in normal cells, is aided by the presence of an enzyme which repairs progressive chromosomal shortening. The lack of chromosomal shortening allows the malignant cells to undergo many more divisions than the normal cells. Which of the following enzymes is most likely to have this effect?

- a. Reverse transcriptase
- b. DNA polymerase
- c. Topoisomerase
- d. Protein kinase
- e. Telomerase

In an experiment, it is observed that chronic, increased exposure to ionizing radiation results in damage to cellular DNA. As a consequence, a protein is now absent that would arrest the cell in the G1 phase of the cell cycle. Subsequent to this, the cell is transformed to acquire the property of unregulated growth. The absent protein is most likely the product of which of the following genes?

- a. RAS
- b. MYC
- c. BCL-2
- d. ABL
- e. TP53

In the equilibrium phase of cancer immunoeediting?

- a. Tumors are less immunogenic
- b. Characterized by presence of minimal residual disease (MRD)
- c. There is continuous elimination of tumor cells
- d. Solely maintained by innate immunity
- e. The shortest of the three phases of cancer immunoeediting

In the escape phase of the cancer immunoeediting

- a. The immune response and tumor growth will be in equilibrium
- b. DCs present tumor antigen in a tolerogenic fashion
- c. MDSCs avail important amino acids such as tryptophan for the tumor growth
- d. Tumor cells induce CD28
- e. There will be no tumor specific CTLs

In which scenarios is multinucleated giant cells not commonly seen?

- a. Malignant neoplastic process
- b. Rhabdomyosarcoma
- c. Tuberculosis
- d. Fibroma
- e. Foreign body reaction

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It is observed that some neoplasms appear to develop from viral oncogenesis, with serologic confirmation of past viral infection. Which of the following neoplasms is most likely to develop in this manner?

- a. T-cell leukemia
- b. Retinoblastoma
- c. Cerebral astrocytoma
- d. Small cell anaplastic carcinoma
- e. Prostatic adenocarcinoma

Mutations of the p53 gene frequently occur in

- a. Huntington's Disease
- b. Bronchial Carcinoma
- c. Cystic Fibrosis
- d. Colonic polyps
- e. Type 2 Diabetes Mellitus

Of the following histopathologic findings, the one that best indicates that a neoplasm is malignant is

- a. Pleomorphism
- b. Necrosis
- c. Increased nuclear/cytoplasmic ratio
- d. Invasion
- e. Atypia

One of the following changes is least likely with familial colonic cancer:

- a. Microsatellite instability
- b. Chromosomal translocation
- c. Loss of heterozygosity for tumor suppressor genes
- d. APC gene mutation
- e. DNA repair gene mutation

One of the following is NOT a mechanism of immune evasion by tumor?

- a. Upregulation of inhibitory checkpoint molecules
- b. Upregulation of costimulatory molecules
- c. Secretion of suppressive molecules
- d. Recruitment of immune suppressive cells
- e. Loss of tumor antigen expression

Patients with partially active/ inactive dihydropyrimidine dehydrogenase (DPD) enzyme are at higher risk for toxicity from:

- a. Nitrosoureas / active/ inactive dihydropyrimidine dehydrogenase (DPD) enzyme are at higher risk for toxicity from.
- b. 6-Mercaptopurine
- c. Methotrexate
- d. 5-Fluorouracil
- e. 6-Thioguanine

Possible explanation for tumor necrosis include:

- a. Incomplete differentiation of less mature cells
- b. Hyperplastic epithelium
- c. Reverse differentiation of mature normal cells
- d. Various mitotic figures
- e. Rapid growth of tumor cells

Radiation therapy destroys:

- a. All cells in the tumor instantly
- b. The cells in the center of the tumor
- c. Primarily rapidly dividing cells
- d. Radioresistant cells

The correct statement about the role of microbes in cancer is:

- a. Most people that are infected with oncogenic pathogens develop cancer
- b. The disruption of human microbiome causes cancer
- c. 15% of cancer infections are a result of oncogenic bacteria
- d. Oncoviruses are responsible for the majority of cancers due to microbes
- e. C & D

The gain of function in RAS gene mutation is equal to the loss of function mutation in which of the following?

- a. Bcl-2
- b. c-myc
- c. GAP protein
- d. Bcr-tyrosine kinase
- e. Rb

The genetic defect involved in HER 2 positive but estrogen receptor negative breast cancer is:-

- a. Point mutation
- b. Amplification
- c. Growth factor binding
- d. Reduced mRNA expression
- e. Epigenetic alteration

The main mechanism of action of Pemetrexed is:

- a. Alkylation of DNA
- b. Inhibition of RNA
- c. Inhibition of thymidylate synthase
- d. Inhibition of DHFR
- e. Free radical generation

The normal nuclear to cytoplasmic ratio is?

- a. 1:2
- b. 1:8
- c. 4:1
- d. 4:2
- e. 1:4

The process of carcinogenesis usually begins with:

- a. Development of defective genes
- b. A single exposure to a known risk factor causing temporary cell damage
- c. An irreversible change in the cell DNA
- d. Exposure to promoters causing dysplasia

The spread of malignant cells through blood and lymph to distant sites is:

- a. Seeding
- b. Invasiveness
- c. Systemic effect
- d. Metastasis

Tumor antigens recognized by T cells:

- a. They are expressed by other normal cells
- b. Can not be targets for immunotherapy
- c. Are tumor specific
- d. There are no tumor antigens recognized by T cells
- e. Can not be used for development of vaccine

Vindesine is different from Vincristine in that it:

- a. Differs in clinical pharmacology profile
- b. Differs in mechanism of action
- c. Differs in mechanism of resistance
- d. Differs in safety profile

What are the common risk factors for the development of breast cancer?

- a. A female sex
- b. Advanced age
- c. Hormone replacement therapy
- d. Family history of breast cancer
- e. all

What are the possible side effects of platinum containing chemotherapy agents?

- a. Nephrotoxicity
- b. Neurotoxicity
- c. Ototoxicity
- d. Cardiac arrest
- e. All
- f. All except D

What is different about Cabazitaxel from other taxanes?

What is different for Cabazitaxel from other taxanes?

- a. It differs in its elimination
- b. It differs in that it is a semi synthetic derivative
- c. It differs in that it is a poor substrate for P glycoprotein efflux pump
- d. It differs in its metabolism
- e. It differs in its mechanism of action

What is the first event after a growth factor /cytokine binds to its cognate receptor?

- a. It causes receptors to dimerize.
- b. It causes tyrosine in the receptor's C-terminal tail to become *phosphorylated*.
- c. It causes the receptor to pick up a tyrosine kinase from the cytoplasm.
- d. It activates the receptor's cytoplasmic tyrosine kinase domain.
- e. It causes the receptors regulatory domain become activated.

What is the role of radiotherapy in cancer treatment?

Which is Not true concerning oxaliplatin?

- a. Myelosuppression is the main dose limiting toxicity
- b. Results in interstrand and intrastrand crosslinks which result in inhibition of DNA synthesis and function
- c. A third generation platinum analog
- d. Has synergistic interaction with other alkylating agents
- e. Tumors that are resistant to cisplatin are not cross resistant to oxaliplatin

Which is True concerning 5 Fluorouracil?

- a. Bolus schedules are preferred for drug administration
- b. It is a prodrug
- c. Cytotoxicity is the result of only DNA synthesis inhibition through "thymineless death."
- d. Administered orally
- e. It has long half life

Which of the following molecules is a cancer antigen presentation stimulating factor?

- a. TGF β
- b. VEGF
- c. LAG-3
- d. CD40/CD40L
- e. PD-L1

Which of the following can be considered as a paraneoplastic syndrome?

- a. Cushing syndrome in small lung carcinoma
- b. Anemia in colorectal cancer
- c. Disseminated intravascular coagulation (DIC) in acute promyelocytic leukemia
- d. Hypoglycemia in insulin secreting B-cell adenoma of pancreas

Which of the following cellular mechanisms is most likely to account for tumor response to radiation therapy?

- a. Secondary inflammation
- b. Loss of the blood supply
- c. Point mutations in DNA
- d. Adenosine triphosphate depletion
- e. Generation of free radicals

Which of the following contribute to the metastatic spread of cancer cells?

- a. Cytoskeletal changes and cell adhesion/motility
- b. Increased enzyme production
- c. Loss of contact inhibition
- d. Pseudopod formation
- e. All of the above are correct.

Which of the following doesn't have autonomous growth?

- a. Hemangioma
- b. Fibroma
- c. Osteblastoma
- d. Cervix
- e. Wilm's tumor

Which of the following factors is most strongly linked to an increased risk for the development of multiple myeloma?

- a. Mustard gas exposure
- b. Alcohol use
- c. Ionizing radiation exposure
- d. Tobacco use
- e. Vinyl chloride exposure

Which of the following hereditary disorders does not predispose to malignancy?

- a. Cystinuria
- b. Ataxia-telangiectasia
- c. Xeroderma pigmentosum
- d. Bloom's syndrome
- e. Fanconi's anemia

Which of the following is a characteristic of malignant neoplasia?

- a. Hemorrhage
- b. Thickened epithelium
- c. Apoptotic bodies
- d. Anaplasia
- e. Mitosis

Which of the following is a wrong pairing

- a. Infections with hepatitis C (HCV) virus increase the risk of hepatocellular carcinoma (HCC)
- b. Epstein-barr virus (EBV) infection causes esophageal adenocarcinoma
- c. Persistent *Helicobacter pylori* infection is associated with mucosa-associated lymphoid tissue (MALT) lymphoma
- d. Infections with *Schistosoma* are linked to bladder cancer
- e. Infections with *Bacteroides* species colon cancer

Which of the following is an immune-suppressive molecule?

- a. prostaglandin E2
- b. IFN α
- c. Granzyme
- d. IFN- γ
- e. IL-2

Which of the following is correct statement about oncoviruses?

- a. Viruses alone are responsible for over 20% of the world's cancers
- b. Oncoviruses can cause genetic instability on infected host cells but not mutation
- c. All the oncoviruses are RNA viruses
- d. Both DNA and RNA viruses can be oncoviruses.
- e. Reverse transcriptase is present in all oncoviruses

Which of the following is incorrect

- a. Up to 20% of human malignancies are related to chronic inflammation and to viral and bacterial infections
- b. Tumor cells do co-opt some of the signaling molecules of inflammatory response
- c. The risk is small but acute inflammation is known to be a risk for cancer
- d. Many environmental causes of cancer and risk factors are associated with some form of chronic inflammation
- e. Chronic inflammation induced by non-infectious agents can contribute to carcinogenesis

Which of the following is matched correctly?

- a. Hepatocellular carcinoma – bile
- b. well differentiated SCC –keratin
- c. Osteosarcoma –osteoid
- d. Small cell carcinoma of the lung – ACTH
- e. Breast Carcinoma –renin

Which of the following is NOT a tumor-promoting immune cell?

- a. Tumor-associated neutrophils (TANs)
- b. Natural killer cells
- c. regulatory or tolerogenic DCs (tDCs)
- d. Myeloid derived suppressor cells
- e. Type 2 macrophages

Which of the following is NOT true about role of the immune system in cancer?

- a. It is possible to target it in the development of a therapy
- b. It controls tumor outgrowth
- c. It has no role in cancer
- d. It shapes tumor immunogenicity
- e. It confers protection against cancer

Which of the following is NOT TRUE about tumor antigens?

- a. Could be of viral origin
- b. May not be involved in survival or dissemination of the tumor
- c. Could be overexpressed
- d. Could be causes for cellular transformation
- e. Could be downregulated

Which of the following is NOT true about role of the immune system in cancer?

Which of the following is NOT TRUE about tumor antigens?

- a. Could be of viral origin
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Which of the following is true about BRCA2?

- a. It is less common than BRCA1 as a risk of breast cancer
- b. Its mutation can be present in sporadic breast cancer
- c. Located on chromosome 17
- d. It is not responsible for breast cancer in men.
- e. It is an oncogene

Which of the following neoplasms occur primarily in childhood?

- a. Osteosarcoma
- b. Neurofibroma
- c. Neuroblastoma
- d. Gliosarcoma
- e. HCC

Which of the following replication and persistence strategies is not characteristic of *oncogenic virus*?

- a. Induction of cell cycle
- b. Metabolic reprogramming
- c. Prevention of latent episomes or provirus from multiplication to arrest cell proliferation
- d. Prevention of apoptosis until viral particles mature
- e. Induction of angiogenesis

Which of the following statements is wrong about the roles of oncogenes during host cell transformation?

- a. The integration of proviruses into host cell genomes near proto-oncogene regulatory sequences can result in host cell transformation.
- b. Insertion of viral oncogenes into host cell genomes can result in host cell transformation.
- c. The alteration of host cell cycle regulation by viruses can lead to host cell transformation.
- d. The permanent activation of host cell signaling pathways by viruses due to chronic infection can result in host cell transformation.
- e. Chronic Hepatitis C Virus can result in liver cell transformation due to the integration of the viral genome into infected liver cells (HCV).

Which one best describes the biomolecular mechanism of oncogenesis by *Candida albicans*?

- a. Production of Acid aspartyl proteinase
- b. Down-regulation in proinflammatory cytokines
- c. Production of acetaldehyde
- d. Reduction of reactive oxygen species
- e. A and C

Which one is a growth factor oncogene?

- a. MYC
- b. JUN
- c. SIS
- d. FOS
- e. RB

Which One is a pyrimidine nucleoside analogue?

- a. Fludarabine
- b. Clofarabine
- c. Cladribine
- d. Cytarabine

Which one is correct about mechanism of action of Etoposide?

- a. Alkylation of DNA
- b. Generation of free radicals
- c. Enhancement of tubulin polymerization
- d. Inhibition of topoisomerase II
- e. Inhibition of tubulin polymerization

Which one is false about tumours?

- a. Most elements in TME work together to promote tumour survival and/or evasion of the immune system.
- b. TME is exclusively constituted of cancer-associated fibroblast (CAFs) and endothelial cells.
- c. Tumour microenvironment (TME) can dictate aberrant cellular function and play a critical role in the subsequent development of more advanced and refractory malignancies.
- d. Tumours are organ-like structures composed of different cell types.
- e. Different cells in a tumour interact to promote maintenance, growth and metastasis.

Which one is Not a component of reactive stroma surrounding a neoplasia?

- a. Cytotoxic T cells
- b. Blood vessel
- c. Natural Killer cells
- d. Connective tissue
- e. Fibroblastic proliferation

Which one is Not an anthracycline?

- a. Bleomycin
- b. Doxorubicin
- c. Daunorubicin
- d. Idarubicin
- e. Epirubicin

Which one is Not correct concerning Alkylating agents?

- a. Result in cross-linking of DNA strands
- b. Result in miscoding through abnormal base pairing with thymine
- c. Are themselves carcinogens in nature
- d. Are potent vesicants
- e. Decreased repair capability to DNA lesions is one of the causes for resistance

Which one is Not correct regarding primary induction treatment of chemotherapy?

- a. Main approach in patients with advanced metastatic disease
- b. Prolong time to tumor progression
- c. This approach has high cure rate
- d. Relieve tumor related symptoms
- e. Confer survival benefit when compared with supportive care

Which one is Not true about tumour cells and their metabolism?

- a. Aerobic glycolysis by tumour cells is intended to produce sufficient amount of ATP and limited number of ingredients for cellular proliferation.
- b. Tumour cells promote their proliferation by reprogramming metabolism to increase glucose uptake and stimulating lactate production.
- c. The rate of glucose uptake dramatically increases and lactate is produced, even in the presence of oxygen and fully functioning mitochondria
- d. Tumour cells tend to favour a specialised fermentation over the aerobic respiration pathway that most other cells of the body prefer.

Which one is Not true concerning Nitrosoureas?

- a. All require enzymatic biotransformation
- b. Are highly lipid soluble
- c. Are effective for treatment of brain tumor
- d. Are alkylating agents
- e. Are non-cross resistant with other alkylating agents

Which one is Not true regarding Methotrexate?

- a. Leucovorin can reverse the effect of methotrexate
- b. Dose modification is required in renal dysfunction
- c. Intracellular polyglutamate form is critically important for its therapeutic action
- d. Aspirin facilitates renal excretion of Methotrexate
- e. Administered by intravenous, intrathecal, or oral route

Which one is True about Aflatoxin-B1?

- a. They are transformed to epoxides by *Aspergillus* species
- b. Are produced by cytochrome P-450
- c. Their epoxide forms react with nucleic acids forming adducts
- d. They cause G → I transition
- e. A and D

Which one of the following cancers is induced by an Aflatoxin B-1 that is derived from a fungus, mostly contaminates grains and taken in by humans?

- a. Hepatocellular carcinoma
- b. Squamous cell carcinoma of skin
- c. Renal cell carcinoma
- d. Adenocarcinoma of rectum
- e. Transitional cell carcinoma of lung